

Ivan Kirischian

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Website: bigbirdy7.github.io/myprojects

About me:

I am a high school student with a passion for robotics, coding, and technology, actively seeking opportunities to teach and mentor younger students. I am always eager to share my experience in designing, building, and programming robots and electronics projects. As a strong communicator and adaptable team player, my greatest soft skill is efficiently explaining complex concepts from the foundation up. To apply this, I run a solo tutoring business teaching math and programming to kids, while also regularly tutoring my peers. I also enjoy volunteering; recently, I ran the junior VEX V5 section for my school's STEM Day, teaching visiting students how to drive the robots.

Skills

- Robotics/Electronics: VEX V5, Arduino, Mbot, Microbit, LEGO Mindstorms
- Programming: Java, Python, C++ and Block-based Coding (E.g. Scratch)
- Experienced with CAD tools like Tinkercad, SketchUp and Fusion360 (for VEX V5)
- Strong communication skills and patience when teaching others. My greatest soft skill is efficiently explaining complex concepts, starting from the foundations.
- I enjoy teaching, and always try to volunteer at positions that allow me to do so. For example I ran the junior VEX V5 section for STEM day for the visiting students, and taught them how to drive the robot.
- I run a solo tutoring business, teaching kids math and programming, while also regularly tutoring my sister and classmates.

Awards

Skills Regional Vex Competition– *[March, 2026]*

- Got Second at the YRSC for VEX V5 robotics, moving onto provincial competition.
- Led the junior team to create a functioning robot in less than 1 week for the schools STEM day

McMaster Sumo Bot Competition: **First Place**– *[January, 2025]*

- Main developer for a robot controller to compete in an autonomous sumo wrestling competition.
- Collaborated with teammates to: build, debug, and optimize the robot operations

School Awards

- Top of the class for Grade 12 Computer Science course (done in grade 11)
- Top of the class for Grade 11 Computer Science course (done in grade 10)

Personal Projects (bigbirdy7.github.io/myprojects)

- **3D Renderer and Flight Sim, From Scratch**
Programmed a 3D renderer in Java, completely from scratch, without copying tutorials. The only libraries it uses are a library for coloring pixels on a window. I learned about linear algebra,

rendering and rasterising, then used my knowledge to create the project. Later I added multiplayer functionality.

- **Robot Car with Laser Turret**

Built a mobile robot with a laser turret controllable by a smartphone app, took apart servos and replaced inner components to achieve our project goal in a limited timeframe.

- **Real Life Portal Turret with AI vision (In-progress)**

A 3D printed turret that tracks a person as they walk through the room and “shoots” at them with LED lights. I greatly improved my soldering skills when making the project..

- **Math Quiz Device for Kids**

Created an educational game using an Arduino, keypad, and LCD to help my younger sister practice math, complete with a point system and a custom 3D-printed case.

- **Personal AI Assistant**

Desktop AI assistant using Groq API, microphone input, and a text-to-speech model. Inspired by the character GLaDOS from Portal 2. I am planning to integrate it into the Real Life Portal Turret.

Education

Thornhill Secondary School – Grade 11 Student

Graduation: 2027

- Relevant Courses: Computer Science 98% (grade 12) Computer Science 97% (grade 11), Computer Engineering 97% (grade 11) Computer Engineering 100% (grade 10)
- Member of Robotics Club (Will be president next year) and Coding Club